

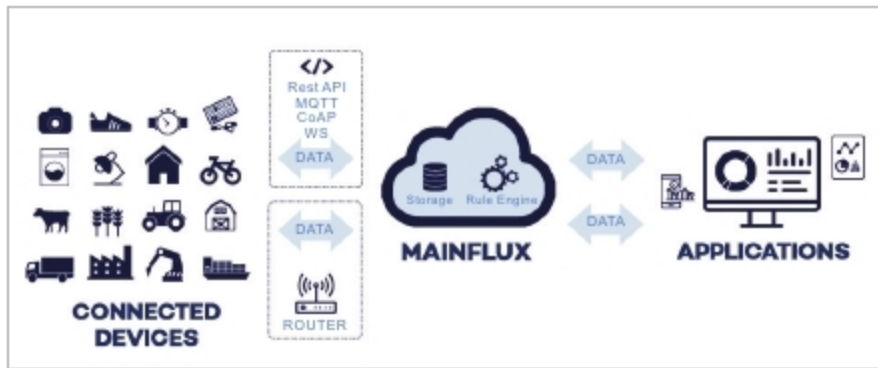


Figure 7: MainFlux

MainFlux

MainFlux is an open source, modern, scalable, secure and patent-free IoT platform based on the cloud. It is completely designed using the Go programming language. It provides a strong platform for the development of IoT solutions, applications and products. MainFlux has powerful capabilities to manage large scale deployments and voluminous transactions. The back-end cloud platform configuration and APIs provide easy device connectivity without any sort of adjustments to enable end users to save time, and provides visualisations and analytical tools on dashboards for quick analysis. Unlike other IoT platforms, this platform supports all protocols—MQTT, WebSockets, CoAP and REST API.

Features

- Supports SDK libraries to help end users to code in C/C++, JavaScript, Go and Python.
- Easy deployment and highly scalable platform using Docker.
- Fine-tuned security using MainFlux authentication and the authorisation server with an access control scheme based on API keys and scoped JWT.
- LoRa integration.
- Supports a lot of hardware and intelligent devices using open source hardware platforms —WeIO is integrated with MainFlux.

Official website: <https://www.mainflux.com/>

ThingSpeak

ThingSpeak is regarded as the topmost open source platform for IoT. It's an IoT application and uses an API to store and retrieve data from things using the HTTP protocol. It facilitates the creation of sensor logging applications, location tracking applications and a social network of things with status updates.

It allows end users to aggregate, visualise and analyse live data streams in the cloud. ThingSpeak has integrated support from numerical computing software like MATLAB, allowing users to analyse, visualise and upload data using MATLAB without having to purchase the software. The platform collects and stores sensor data in the cloud and develops IoT applications. It fully supports a range of development boards like Arduino, ESP8266, BeagleBone and Pi.

The core element of ThingSpeak is a 'ThingSpeak Channel'. This stores the data that we send to ThingSpeak and

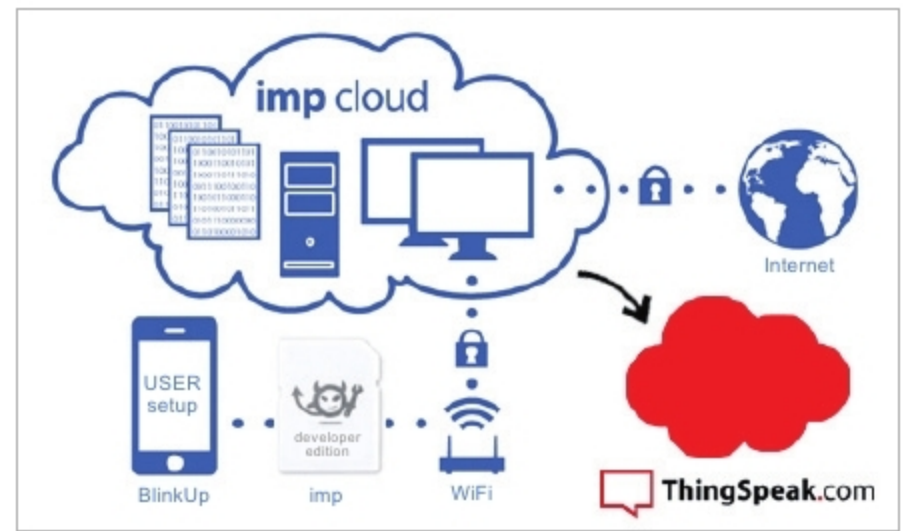


Figure 8: ThingSpeak

comprises the following:

- Eight fields for storing data of any type — these can be used to store the data from a sensor or from an embedded device.
- Three location fields to store the latitude, longitude and the elevation. These are very useful for tracking a moving device.
- One status field, which is a short message to describe the data stored in the channel.

To use ThingSpeak, one needs to sign up and create a channel. Once that is done, send the data, and allow ThingSpeak to process and retrieve it.

Features

- Easy device configuration to transmit data to the ThingSpeak platform using standard IoT platforms.
- Real-time visualisation of sensor data and data aggregation using third party sources.
- IoT analytics runs based on schedules or events.
- MATLAB analytics, RESTful and MQTT APIs.
- Enables the prototyping and building of IoT systems without any server setup and Web software installation.
- Automatically acts on data and communication via third party services like Twilio or Twitter.

Official website: <https://thingspeak.com/> 

References

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